

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative modeler with experience in stochastic processes, approximate likelihoods, sequential testing, and large-scale risk modeling. Builds scalable Python/PyTorch systems and model diagnostics for dependent data on AWS and HPC.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, PyTorch, LightGBM, scikit-learn), R, SQL/MySQL, Linux/HPC, AWS EC2

Quantitative Methods: Stochastic processes, MLE, Gaussian processes, approximate likelihood, spectral methods, sequential testing, model validation, GLM/GBM

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Designed a sequential anomaly-detection framework for binary risk indicators by representing cumulative evidence as a random walk and using Brownian-motion approximations to quantify stopping-time uncertainty and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline for mid-market property risk - data processing, GLM benchmarking, validation, and diagnostics - across 2.46M+ records on AWS EC2.
- Translated model diagnostics and feature-attribution patterns into pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND PROJECTS

Ph.D. Dissertation Research - Scalable Gaussian Process Inference and Diagnostics for Spatiotemporal Data

Sep 2024 - Present

- Scalable Inference:** Developed an advection-aware Vecchia likelihood for nonseparable spatiotemporal Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- Model Diagnostics:** Designed scale- and frequency-resolved diagnostics to identify misspecification in covariance smoothness, noise, spatial and temporal dependence, and space-time interactions.
- Gaussian Process Modeling:** Built and compared Matérn and generalized Cauchy covariance models that capture multiscale dependence, temporal evolution, and advection.
- Spectral and Physical Validation:** Derived missing-data-aware spectral diagnostics and used directional cross-variograms to evaluate advection and wind-direction consistency.
- Research Engineering:** Implemented estimation, simulation, and diagnostic pipelines in Python/PyTorch with CPU/GPU execution and restartable HPC workflows.

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance evaluation.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)